

Academic curriculum vitae

Shirin Hajahmadi

Position: Postdoctoral Researcher, Department of Computer Science and Engineering,
University of Bologna

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Google Scholar: Google Scholar Profile

ORCID link: ORCID Profile

Web: Personal Website | University Profile



Profile Summary

Postdoctoral Researcher in Human–Computer Interaction, specializing in cognitive augmentation with Extended Reality (XR), Artificial Intelligence (AI), and sensor systems. Publications: IEEE VR, ISMAR, VRST, ACM SUI; journals: *Frontiers in Virtual Reality*, *Sensors*, *Digital Applications in Archaeology and Cultural Heritage*. Experienced in leading interdisciplinary teams and building XR tools for users with diverse cognitive and motor abilities. Funding: NextGenerationEU PNRR; University of Bologna Marco Polo, Institute of Advanced Studies (ISA), and Ph.D. fellowship; Emilia–Romagna Region fellowship; CSTC (Iran) workshop grant.

Research Interests

- Human–Computer Interaction and Human–AI Interaction, with emphasis on developing XR applications that integrate intelligent systems and sensor–based technologies to support cognitive augmentation.
- Security and privacy in extended reality applications, with emphasis on digital identity protection and the ethical design of interactive systems.
- Inclusive and diversity–aware design of interactive systems.

Three Selected Publications

- Hartfill, J., **Hajahmadi, S.**, Schmidt, S., Marfia, G., & Steinicke, F. (2025). Embracing differences in virtual reality: Inclusive user-centered design of bimanual interaction techniques. *Frontiers in Virtual Reality*, 6, 1586875. <https://doi.org/10.3389/frvir.2025.1586875>
- **Hajahmadi, S.**, Cascarano, P., Mostajeran, F., Heuer, K., Lux, A., Mends–Cole, G. O., Steinicke, F., & Marfia, G. (2025, March). Investigating the impact of voice-only and embodied conversational virtual agents on mixed reality puzzle solving. In *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR 2025)*. IEEE. <https://doi.org/10.1109/VR59515.2025.00083>
- **Hajahmadi, S.**, Stacchio, L., Giacché, A., Cascarano, P., & Marfia, G. (2024, October). Investigating extended reality–powered digital twins for sequential instruction learning: The case of the Rubik’s Cube. In *Proceedings of the IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2024)* (pp. 259–268). IEEE. <https://doi.org/10.1109/ISMAR62088.2024.00040>

Education and Training

**University of Bologna, Department of Computer Science and Engineering /
VARLab**

Bologna, Italy

Ph.D. in Computer Science

Nov. 2021 – Apr. 2025

- **Thesis title:** Cognitive Augmentation Systems through Extended Reality, Artificial Intelligence, and Sensors: A Human–Computer Interaction Perspective

Institute for Cognitive Science Studies / Shahid Beheshti University

Tehran, Iran

M.Sc. in Cognitive Science (Cognitive Psychology)

Oct. 2016 – Jan. 2019

- **Thesis title:** Map-based alignment effects on virtual spatial navigation in complex curved layouts

Shahid Bahonar University

Kerman, Iran

B.Sc. in Computer Science

Feb. 2006 – Apr. 2010

- **Thesis title:** Toward the design and development of lexical and syntactic compilers

Tutoring & Mentoring

eXRAI: Exploring XR and AI in the Creative Industries Workshop (sponsored by University of Bologna and Universidad Rey Juan Carlos)

Bologna, Italy

Workshop Lecturer

Jul. 2024

- Taught Unity for creating virtual environments and supervised the implementation of an example Unity project (6 hours).

VR/AR Laboratory (MSc Computer Science) — Department of Computer Science and Engineering, University of Bologna

Bologna, Italy

Tutor (instruction-only)

2019–2025

- Led tutorials and labs; weekly office hours; project mentoring; no grading responsibilities.
- 75–100 students total; 20 contact hours.

Data Science and Immersive Technologies for Fashion E-commerce (MSc Fashion Studies) — Department of Arts, University of Bologna

Rimini, Italy

Tutor (instruction-only)

2020–2024

- Led tutorials and labs; weekly office hours; project mentoring; no grading responsibilities.
- 40–45 students total; 20 contact hours.

Advanced Information and Communication Technologies (MSc Fashion Studies) — Department of Arts, University of Bologna

Rimini, Italy

Lecturer

2023–2024

- Led tutorials and labs; weekly office hours; project mentoring; no grading responsibilities.
- 3 students total; 18 contact hours.

Supervision Experience

Department of Informatics, University of Hamburg

Hamburg, Germany

Project Supervisor, BSc Lab Course

2023–2024

- Supervised student teams on social dynamics and task performance in mixed-reality puzzle activities under different virtual-assistant conditions.
- **Video demo; Publications:** SUI 2024 poster, IEEE VR 2025.

Department of Computer Science & Department of Fashion Studies, University of Bologna

Bologna/Rimini, Italy

Co-supervisor (MSc theses)

2022–2023

- Maryam Arjomandi — *VR fashion experience inspired by “Alice in Wonderland”*. [Link to the description](#).
- Igor Iurevici — *User enactments in VR: Smart home interaction study*.
- Ranim Elnabouche — *Virtual fashion show as a gateway to avatar customization*. **Video demo; Publication:** IEEE AIXVR 2025 Workshop.
- Yue Chen — *Retail-metaverse synergy for consumer engagement*.

University of Bologna – Department of Fashion Studies

Rimini, Italy

Course Project Supervisor (MSc Fashion Studies) – [Projects description link](#)

2021–2022

- The Antique Museum — Mina Etesam. Two-room virtual museum (lounge + gallery) with interactive exploration. [GitHub link](#)
- Thierry Mugler Virtual Exhibition — Meizi Yu; Ting Su. Tribute exhibition with curated assets and narrative flow. [GitHub link](#)
- Ralph Lauren F/W 2022 Collection — Afshan Khan. Virtual showroom with brand timeline and inspiration board. [GitHub link](#)
- Urban Glamour in Gucci Style — Chiara Melchiori; Elizaveta Uvarova. City-themed showroom with try-on interactions.

Positions held to date

Department of Computer Science and Engineering, University of Bologna

Bologna, Italy

Postdoctoral Researcher

Apr. 2025 – Present

- Conducting research in security and privacy paradigms in Extended Reality (XR), with responsibilities for system design, implementation, and empirical evaluation.

Human–Computer Interaction Lab, Department of Informatics, University of Hamburg

Hamburg, Germany

Visiting Researcher

Sep. 2023 – Nov. 2023

- Collaborated on the design, development, and evaluation of inclusive Virtual Reality (VR) for users with limb loss, and supervised the design and development of a Mixed Reality (MR) assistant for puzzle-solving.

Dept. of Computer Science and Engineering (VARLab), University of Bologna

Bologna, Italy

Doctoral Researcher

Nov. 2021 – Apr. 2025

- Conducted research on XR, AI, and sensors for cognitive augmentation.

University of Bologna & Anastasis Srl — COMPRENDO Project

Bologna, Italy

Research Fellow

Mar. 2020 – Oct. 2021

- Collaborated on the design and development of inclusive AR and desktop applications aimed at cognitive augmentation for children with autism.

Shahid Beheshti University

Tehran, Iran

Research Assistant

Feb. 2019 – Mar. 2020

- Conducted studies on spatial cognition in virtual environments.

Fellowships & Grants

European Union – NextGenerationEU, University of Bologna

Bologna, Italy

PNRR Research Fellowship (Postdoctoral)

2025 – ongoing

- Research on the design, analysis, and implementation of security and privacy paradigms in extended reality (XR) scenarios.

University of Bologna

Hamburg, Germany / Bologna, Italy

Marco Polo Incentive Fellowship

2023–2024

- International research stay at the University of Hamburg (Department of Informatics).

University of Bologna, Department of Computer Science and Engineering / VARLab

Bologna, Italy

Ph.D. Scholarship

2021–2024

- Doctoral research in Computer Science with a focus on cognitive augmentation through XR, AI, and sensors.

Institute of Advanced Studies (ISA), University of Bologna

Bologna, Italy

ISA Fellowship (International Ph.D. College)

2021–2024

- Interdisciplinary research collaborations and community engagement.

Emilia-Romagna Region

Bologna, Italy

Research Fellowship

2020–2021

- Research on the design, development, and implementation of Augmented Reality and desktop applications for inclusive education.

- Organizing and leading a workshop series on task design in spatial cognition studies with VR.

Research Collaborations / Partners

Collaboration on the Sora project: Investigating audiovisual elements and realism in generative AI videos.

Madrid, Spain

Universidad Carlos III de Madrid (Spain)

2024

Collaboration on inclusive VR for users with limb loss and an MR assistant for puzzle-solving project.

Hamburg, Germany

University of Hamburg, Department of Informatics

2023

Collaboration on the Development of an Augmented Reality (AR) + ChatGPT mobile application for inclusive language learning.

Tarragona, Spain

Universitat Rovira i Virgili, Department of Romance Philology

2023

Commissions of Trust and Service

- **Workshop Co-chair & Organizer**, GenAI-XR: Generative Artificial Intelligence meets Extended Reality, IEEE VR 2025, Saint-Malo, France, 8–12 March 2025. [link](#)
- **Technical Program Committee (TPC) Member**, Generative AI and LLMs for XR (GenAI-LLM-XR), IEEE AIXVR 2025 Workshop, Instituto Superior Técnico, Lisbon, Portugal, 27–29 January 2025. [link](#)
- **Reviewer**, Taylor & Francis journals in hypermedia, multimedia, and human–computer interaction (2025).
- **Reviewer**, Frontiers in Computer Science — Human–Media Interaction section (2025).
- **Ad-hoc Reviewer**, ACM Symposium on Virtual Reality Software and Technology (VRST), 2025
- **Ad-hoc Reviewer**, Workshops at IEEE VR, ISMAR, and AIXVR (multiple years, 2023–2025)

Seminars and Conference Presentations

- **IEEE VR 2025 – The 32nd IEEE Conference on Virtual Reality and 3D User Interfaces**: Investigating the Impact of Voice-only and Embodied Conversational Virtual Agents on Mixed Reality Puzzle Solving (Saint-Malo, France, March 2025).
- **IEEE AIXVR 2025 – XRiM Workshop**: Redefining Fashion: A VR Journey from 360 Runway Show to Contextual Avatar Customization (Lisbon, Portugal, January 2025).
- **CREA(I) 2024**: The influence of audiovisual elements on the realism of generative AI videos: The case of Sora (Santiago de Compostela, Spain, October 2024).
- **GRLMC Seminar 2023**: ARELE-bot: Inclusive ELE Learning with AR and ChatGPT (Tarragona, Spain, December 2023).
- **Digital Research in the Humanities and Arts (DRHA) 2023**: Exploring Symbolic Narratives in Virtual Spaces (Turin, Italy, September 2023).
- **Human-Computer Interaction International (HCII) 2023**: Toward a Shared Experience of Uncertainty of Interpersonal Communication Through an Immersive VR Serious Game (Copenhagen, Denmark, July 2023).

Skills

RESEARCH AND METHODS

- Design and execution of qualitative and quantitative user studies, usability testing, interviews, and surveys.

- Experience with cognitive measurements (EMG, motion capture, and eye-tracking) for HCI and XR evaluation.
- Academic writing, publishing, and grant proposal preparation.

TEACHING, MENTORING, AND SUPERVISION

- Supervision of MSc and BSc theses across computer science, cognitive psychology, and fashion studies.
- Mentoring interdisciplinary student teams and leading collaborative research projects.
- Delivery of lectures and workshops in XR, AI, and HCI.

TECHNICAL AND DEVELOPMENT SKILLS

- XR development: Unity, Unreal, ARFoundation, MRTK, SteamVR, OpenXR; hardware including HoloLens, Oculus, HTC Vive, and Varjo. Sensors: optical/IMU motion capture, eye tracking (Tobii/Varjo), heart-rate, EMG, depth cameras (Azure Kinect), hand tracking (Leap Motion), Vive/Quest trackers.
- Programming: Python (data analysis, ML/DL), C# (Unity), C++, Java, Kotlin, Arduino, HTML.
- Deep Learning: Design, training, and optimization of neural networks (CNNs, RNNs, Transformers) using PyTorch/TensorFlow.
- Statistical Analysis: evaluation metrics, and statistical modeling with Python and R.

DESIGN AND PROTOTYPING

- UX/UI design and interaction design for XR applications.
- Prototyping with Figma, SketchUp; media editing with Kdenlive and Audacity.

References available upon request.